SkyPass: A Weather-Aware Integrated Satellite Transit Planning and Observation System

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Abstract—Planning an observation at a small ground station today means stitching together separate tools: one predicts passes, another judges optical visibility, a weather site is consulted by eye, and clashes between overlapping passes are resolved by hand. The weather step is the weakest link, because conventional predictors list passes without regard to cloud cover, and at a cloudy site most listed passes are unobservable. We present SkyPass, an integrated planner that carries a catalogue of orbital elements through propagation, pass extraction, physical visibility scoring, cloud-forecast integration and exact conflict resolution to an executable timetable. Pass extraction uses coarse bracketing with bisection refinement, reaching 0.055 s mean agreement with an independent implementation over 1761 passes while using 23.7× fewer propagator evaluations than a dense scan. Conflict resolution is formulated as weighted interval scheduling under a nightly observing budget and solved by an $O(nK)$ dynamic program that matched exhaustive search on all 2 000 randomised instances tested, where the elevation-chasing rule an operator applies by hand attains 95.39% of the optimum on average and as little as 9.4% in the worst case. We evaluate the weather claim retrospectively at seven stations spanning arid to equatorial-monsoon climates over 60 days, scoring every timetable against ERA5 reanalysis. The central finding is that the value of a forecast is set by the decision it is wired into, not by the forecast alone. A station obliged to observe every night gains little from one — cloud is autocorrelated across an observing session, so a forecast can barely discriminate within it — while a station free to choose which nights to spend its limited effort on gains several times more. Under a window budget SkyPass raises realised yield by 37.6% and lifts the share of scheduled passes falling under clear sky from 13.9% to 23.1%, recovering 35% of what perfect foresight would deliver; tightening the budget further raises the gain to 55.0%. We also show that the raw forecast must not be used directly: numerical cloud fields are strongly overconfident, and feeding them to the scheduler unchanged is worse than ignoring weather altogether. SkyPass is open source and every reported number is regenerated from the released scripts.

Index Terms—satellite pass prediction, SGP4, ground station scheduling, interval scheduling, weather forecasting, forecast calibration, optical observation

I. Introduction

A small ground station — a university laboratory, an amateur-radio club, an optical tracking hobbyist — has a recurring planning problem. Several hundred satellites are potentially observable; over a week they generate thousands of passes; only a few dozen are worth the operator's time; and the station can attend to one object at a time.

The tools available address fragments of this problem. Pass predictors such as Gpredict, Heavens-Above and n2yo compute rise and set times from two-line elements (TLEs) and the SGP4 propagator [1], [2]. Some add an illumination test so that optical observers are not shown daylight passes. A few consumer applications display a weather icon alongside a pass, but we are not aware of one that lets the forecast rank passes or drive a scheduling decision; and none of these tools resolve conflicts between overlapping passes. The operator is handed a list and left to reconcile it by hand.

The weather omission is the consequential one. At the site used as our running example (Chennai, India, 12.92° N) the mean observed cloud cover over optically viable passes in our evaluation window was 88% — monsoon conditions under which a majority of geometrically excellent passes are simply not observable. A predictor that ranks passes by culmination elevation will confidently recommend an overcast night.

The scheduling omission is subtler but equally real. Once several satellites are tracked, their passes overlap, and a single-aperture station must choose. This is a genuine combinatorial problem, and the literature that treats it seriously does so from the operator's side of the link: spacecraft tasking and satellite range scheduling, formulated as mixed-integer programs or attacked with metaheuristics [3]–[7]. Those formulations carry constraints a small station does not have (multi-antenna resources, energy and memory budgets, downlink contention) and solver dependencies it cannot justify.

This paper argues that the small-station problem has a much simpler exact structure, and that closing the weather gap is what actually changes outcomes. Our contributions are:

1) An integrated pipeline that takes a station and a horizon to an executable timetable in one pass, with per-stage accounting of how many passes each physical constraint removes (Section III).

2) An efficient and validated pass extractor: coarse bracketing plus bisection, agreeing with an independently written implementation to 0.055 s mean while spending 23.7× fewer propagator calls than dense sampling (Section V-A).
3) Exact capacity-constrained conflict resolution. We show that the small-station problem is weighted interval scheduling under a cardinality budget, give an $O(nK)$ dynamic program, and verify optimality exhaustively. No solver, and no metaheuristic, is required (Section III-E).
4) A quantified, verified weather claim. We verify cloud forecasts against ERA5 reanalysis [8], and evaluate the resulting timetables against what the sky actually did, at seven stations across distinct cloud climates (Sections V-C–V-D).
5) Two negative results that matter. Weather-awareness buys little under a fixed nightly quota — a small fraction of what the same forecast delivers once the station may choose its nights — and it is actively harmful when the raw forecast is used uncalibrated. Both are quantified, both are easy to get wrong, and both change how such a system should be built.
6) A measured element-set error budget. Using historical element sets we measure how pass timing decays with epoch age over 33 223 aged comparisons, which fixes where the accuracy actually comes from and yields a concrete staleness threshold (Section V-B).

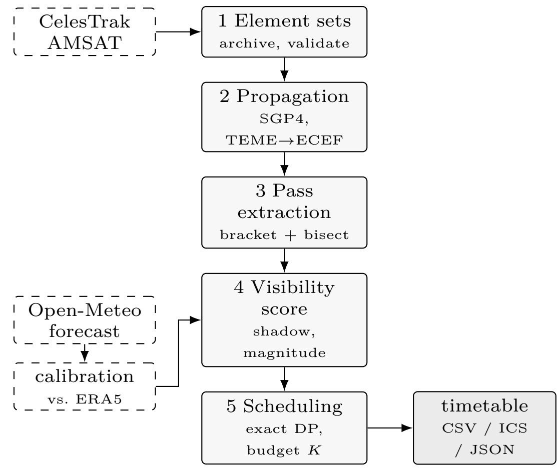

Fig. 1. SkyPass pipeline. The cloud forecast is calibrated against reanalysis before it is allowed to influence the score (Section III-D); the scheduler then resolves conflicts exactly under the station's nightly observing budget.

II. Related Work

A. Pass prediction

SGP4 and its element sets are the standard basis for satellite pass prediction [1]; Vallado et al. republished the reference implementation with corrections [2]. Accuracy is limited by the element sets rather than the propagator: TLE positional error grows from roughly a kilometre at epoch to tens of kilometres after a week [9], [10]. Consumer predictors expose rise/set/culmination times computed by dense sampling or by root-finding on the elevation function; the refinement strategy is rarely reported or validated.

B. Optical visibility

Whether a satellite is seen depends on illumination and photometry, not only geometry. Standard treatments use a conical Earth-shadow model [11] and a diffuse-sphere phase function to convert an intrinsic magnitude into an apparent one [12]. Recent megaconstellation brightness work has refined these models [13], [14] and made visibility prediction newly relevant to observers.

C. Satellite and ground-station scheduling

Satellite range scheduling is NP-hard in general [15] and is usually approached with integer programming or metaheuristics [3]–[6], [16]. Agile Earth observation scheduling is a large adjacent literature [7], [17], [18]. Cloud cover does appear there — as an uncertainty to be hedged in spacecraft imaging tasking [19], [20] — but always on the operator side, deciding what a satellite should photograph, with solver machinery a small station cannot deploy. The observer-side problem, where a fixed station on the ground chooses which passes to watch under its own local sky, has not received the same treatment.

Where the underlying structure is single-machine interval selection, however, exact polynomial algorithms exist: weighted interval scheduling is a textbook dynamic program [21], [22], and the interval-scheduling family is well surveyed [23], [24]. Our contribution here is not the recurrence but the observation that the small-station problem is this problem once a capacity budget is added, together with the extension and its empirical verification.

D. Weather-aware observatory scheduling

Professional observatories do schedule around weather, typically with queue schedulers that re-plan as conditions change [25]–[28]. That work informs our design, but it assumes an institutional software stack and on-site sky monitoring. We target a station whose entire weather input is a free public forecast API.

Forecast verification and the economic value of forecasts are mature fields [29]–[31], and calibration is understood to be essential before a forecast drives a decision [32]. To our knowledge this machinery has not previously been applied to ground-station pass planning; doing so is what turns a plausible idea into a measurable gain.

III. System Design

SkyPass is a five-stage pipeline (Fig. 1). Each stage is a pure function of the previous one, so any stage can be ablated — which is how the weather claim is tested.

Algorithm 1 Pass extraction for one object

Require: element set σ, window $[t_0, t_1)$, mask ε_m, tolerance δ

- $h \leftarrow \max(5, \min(\Delta_{\max}, P(\sigma)/240))$ ▷ adaptive coarse step
- $t \leftarrow t_0$; $e_{\text{prev}} \leftarrow \varepsilon(\sigma, t)$; $a \leftarrow$ nil
- **while** $t < t_1$ **do**
  - $t' \leftarrow \min(t + h, t_1)$; $e \leftarrow \varepsilon(\sigma, t')$
  - **if** $e_{\text{prev}} < \varepsilon_m \le e$ **then**
    - $a \leftarrow \text{Bisect}(t, t', \varepsilon_m, \text{rising}, \delta)$
  - **else if** $e_{\text{prev}} \ge \varepsilon_m > e$ **and** $a \ne$ nil **then**
    - $l \leftarrow \text{Bisect}(t, t', \varepsilon_m, \text{setting}, \delta)$
    - $c \leftarrow \text{GoldenMax}(a, l, \delta)$ ▷ culmination
    - emit pass (a, c, l); $a \leftarrow$ nil
  - **end if**
  - $t \leftarrow t'$; $e_{\text{prev}} \leftarrow e$
- **end while**

A. Element-set management

Element sets are fetched from CelesTrak [33], validated by TLE checksum, deduplicated per NORAD identifier keeping the newest epoch, and archived under the date of retrieval. Archiving matters for reproducibility: CelesTrak serves only current elements, so without a local archive a published prediction cannot be repeated. Sets whose epoch age exceeds $\tau_{\max}$ (default 14 days) are rejected.

B. Propagation and pass extraction

Positions come from the reference SGP4 implementation. The TEME vector is rotated to ECEF by Greenwich mean sidereal time [34] and converted to topocentric azimuth, elevation and range on the WGS-84 ellipsoid. Polar motion is neglected: below 15 m at the surface, far under the SGP4 error budget. Refraction [35] is reported but not applied to the horizon mask, so results do not depend on assumed surface conditions.

Because cost is dominated by propagator calls, pass extraction is designed to minimise them (Algorithm 1). Elevation is sampled on a coarse grid to bracket horizon crossings; within a pass elevation is smooth and unimodal, so a sign change of $\varepsilon(t) - \varepsilon_{\text{mask}}$ between adjacent samples brackets exactly one crossing. Each bracket is then refined by bisection to 0.1 s, and culmination is found by golden-section search. The coarse step is tied to the orbital period ($P/240$, capped at 30 s) rather than fixed, so that higher orbits are sampled proportionally more cheaply without risking missed transits.

C. Visibility scoring

Each pass receives a score $S \in [0, 1]$ combining a geometric quality term with multiplicative gates:

$$S = \underbrace{(w_e f_e + w_d f_d)}_{\text{geometry}} \cdot f_m \cdot f_s, \tag{1}$$

where $f_e = \min(\varepsilon_{\max}/60°, 1)$ and $f_d = \min(T/480\,\text{s}, 1)$ saturate the culmination elevation and pass duration, f_m is a photometric factor and f_s a sky-condition factor.

The multiplicative form is deliberate. A pass that fails a hard requirement — in eclipse, sky still bright, overcast — must score zero however good its geometry is. An additive score cannot express that, and it is exactly where weather-blind planners go wrong: they let excellent geometry outvote an unusable sky.

For optical work the object must be sunlit and the observer in darkness. Earth shadow uses the dual-cone construction [11]; a cylindrical shadow, common in amateur predictors, misplaces eclipse entry by several seconds in low Earth orbit. Apparent magnitude follows the standard diffuse-sphere relation [12], [36]

$$m = m_{\text{std}} + 5 \log_{10} \frac{d}{1000\,\text{km}} - 2.5 \log_{10} F(\psi), \tag{2}$$

with phase function $F(\psi) \propto (\pi - \psi)\cos\psi + \sin\psi$ normalised to $F(0) = 1$, and f_m maps m linearly onto $[0, 1]$ between a bright reference and the limiting magnitude. In radio mode the illumination and photometric terms are dropped: a radio pass works in daylight and through cloud.

D. Weather integration and calibration

The sky factor is $f_s = 1 - \hat{c}$, where $\hat{c}$ is the expected cloud fraction at culmination, interpolated linearly between hourly forecast samples (snapping to the containing hour biases a six-minute pass by up to half an hour).

Critically, $\hat{c}$ is not the raw forecast. A numerical model reports cloud cover as a deterministic field, but as a predictor it is heavily over-confident. We therefore fit

$$\hat{c} = \mathbb{E}[c \mid c_f] = a + b\, c_f \tag{3}$$

by least squares on a window that ends before the period being planned, restricted to local night hours — the only regime in which an optical planner applies it, and one with a distinct cloud regime and model bias from daytime. The slope b is the honest strength of the signal: $b \approx 1$ means the forecast can be taken at face value, $b \approx 0$ means it carries little information and the calibrated value collapses toward site climatology, which is the correct fallback. Section V-D shows that omitting this step makes weather-awareness actively harmful.

E. Conflict resolution

A single-aperture station observes one object at a time and needs a setup gap g to slew and reconfigure between targets. Selecting the best mutually compatible subset of passes is therefore weighted interval scheduling. Writing $v_j = S_j \pi_j$ for the value of pass j (score times mission priority), sorting passes by finish time, and letting $p(j)$ be the latest pass compatible with j, the textbook recurrence [21] gives the exact optimum in $O(n \log n)$.

That formulation, however, misses what makes the problem interesting. Left unconstrained, a station simply

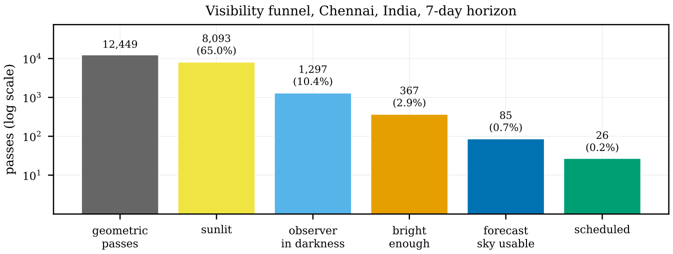

Fig. 2. Visibility funnel for the demonstration station: of 12449 geometric passes over 7 days, 26 survive to the timetable. Each constraint is physical, and each is applied in the pipeline rather than left to the operator.

observes every candidate: in our data an unconstrained planner accepts nearly every non-overlapping pass, so weather-awareness has nothing to choose between. Real small stations are limited not by conflicts but by operator and instrument time. We therefore add a capacity budget of K observations per night and extend the recurrence with a cardinality dimension:

$$\mathrm{OPT}[j][k] = \max\bigl(\mathrm{OPT}[j{-}1][k],\ v_j + \mathrm{OPT}[p(j){+}1][k{-}1]\bigr). \tag{4}$$

This is exact and runs in $O(nK)$ after the sort. Observing nights are disjoint in time, so applying the budget per night and solving each night independently remains exact. Nights are defined in local solar time so that an evening and the small hours that follow belong to the same session.

IV. Experimental Setup

Catalogue. 635 unique objects fetched from CelesTrak GP groups (stations, visual, weather, amateur, science, resource, cubesat, engineering) on the archive date, with checksum validation and per-object deduplication.

Stations. Seven sites chosen to span cloud climatologies: Chennai and Bengaluru (monsoon), Leh (high-altitude arid), Tucson (desert), Singapore (equatorial), Reykjavík and Svalbard (maritime polar). Unless stated, the elevation mask is 10°, the dark-sky threshold −6° (civil twilight), and the setup gap 5 min.

Weather data. All three products come from Open-Meteo [37]: the operational forecast; archived past forecasts at lead times of 1–7 days, which let us score forecast skill without waiting; and ERA5 reanalysis [8] as verification truth.

One methodological caution deserves emphasis, because it silently invalidates the whole evaluation. The archive endpoint must be pinned to the ERA5 model explicitly. At its default setting it returns a best-match series that, for recent dates, is the archived forecast; verifying a forecast against it scores the forecast against a copy of itself and reports a perfect and meaningless skill of 1.0. We observed exactly this before pinning the model. ERA5 is produced by a separate assimilation cycle and is the only series here genuinely independent of the forecast under test.

TABLE I
Pass-timing agreement over the full catalogue (635 objects, 1761 matched passes, 24 h horizon). Values in seconds.

	vs. 1 s dense scan		vs. Skyfield	
Quantity	mean	p95	mean	p95
$\lvert\Delta t_{\mathrm{AOS}}\rvert$	0.504	0.954	0.060	0.157
$\lvert\Delta t_{\mathrm{TCA}}\rvert$	0.248	0.476	0.055	0.110
$\lvert\Delta t_{\mathrm{LOS}}\rvert$	0.502	0.956	0.059	0.156
$\lvert\Delta\varepsilon_{\max}\rvert$ (deg)	0.0008	0.0022	0.0034	0.0102

Reference implementations. Timing is checked against two independent references: a brute-force 1 s dense scan, and Skyfield [38], which shares only the certified SGP4 core and implements time scales, frame transformation and event finding independently. Agreement with Skyfield therefore validates the entire chain downstream of the propagator.

Environment. Python 3.13.5, sgp4 2.27, on a commodity laptop. All runtimes are single-threaded and include no GPU or parallelism.

V. Results

A. Pass-prediction accuracy and cost

Table I compares SkyPass against both references over 635 objects and 1761 matched passes in a 24-hour window.

Against Skyfield, mean agreement is 0.060 s on acquisition and 0.055 s on culmination. This is the meaningful accuracy figure. The apparent 0.504 s "deviation" from the dense scan is not SkyPass error at all: a 1 s grid locates a crossing only to within its own step, so it lags the true crossing by up to a second, with a mean of 0.5 s. The refined estimator is more accurate than the brute-force reference it is conventionally compared against — an artefact worth flagging, since dense sampling is often treated as ground truth.

The cost difference is large: 2 311 148 propagator evaluations against 54 864 000, a factor of 23.7. Detection recall is 99.78% (4 passes missed out of the dense scan's total), the residual being very short grazing transits that peak barely above the mask; lowering the coarse step recovers them at proportional cost.

B. How far do the element sets limit prediction?

Sub-second numerical agreement is only worth having if the orbital elements behind it are that good. In the operational catalogue the median element-set age was 0.42 days (90.2% under one day, maximum 13.04 days). Comparing predictions from two independently curated providers (CelesTrak and AMSAT) for 52 objects present in both, across 308 matched passes, gives a mean culmination difference of 0.041 s (p95 0.176 s); see Table II. Contemporaneous element sets therefore agree to well under a second.

That comparison bounds the disagreement between element sets of the same age; it says nothing about how

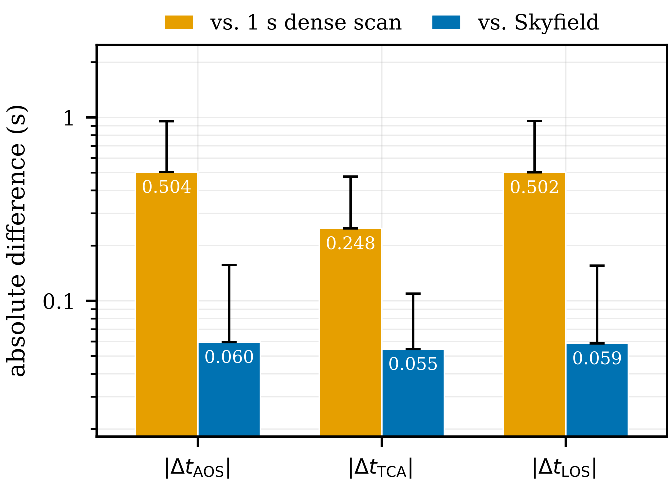

Fig. 3. Pass-timing agreement against both references. Bars are means and caps the 95th percentile; note the logarithmic axis, since the two references differ by roughly an order of magnitude. The larger apparent deviation from the 1 s dense scan reflects that reference's own grid quantisation, not error in the refined estimator.

TABLE II
Pass-timing divergence between two independently curated element sets (CelesTrak vs. AMSAT) for the same objects, grouped by the separation of their epochs.

Epoch separation	N	mean (s)	median (s)	p95 (s)
<0.5 d	279	0.038	0.000	0.149
0.5-1 d	29	0.071	0.022	0.318

prediction decays as elements grow stale. To measure that we need two element sets for the same object at different epochs, with the fresher one as truth, which a single catalogue snapshot cannot provide. We therefore pulled historical element sets from Space-Track's gp_history class for 60 low-Earth-orbit objects, took the freshest available set before a target night as the reference orbit, and re-predicted the same passes from progressively older sets of the same object — 33 223 aged comparisons in all (Table III).

The result completes the error budget (Fig. 4). Within a day of epoch the elements contribute 0.05 s of culmination error, which is the same order as the 0.055 s numerical agreement of Section V-A: at that age the pipeline is balanced, and neither term dominates — the refinement effort of Section III is neither wasted nor masked. Degradation is then sharply super-linear: 4.3 s mean at 7–14 days, 16.2 s at 14–30, and 75.3 s beyond a month. A linear fit through the bin means gives 1.29 s per day, a figure we quote only to note that it badly understates the tail.

This carries a concrete design consequence. SkyPass rejects element sets older than a configurable threshold, and our default of 14 days is too permissive: at 7–14 days the mean culmination error is already 4.3 s, enough to miss the start of a short pass. The measurement argues for a

TABLE III
Degradation of predicted culmination time with element-set age, measured against a fresh reference orbit for the same object (60 LEO objects, 33 223 aged comparisons, historical elements from Space-Track). This is the error the elements contribute, on top of the sub-second numerical agreement of Table I.

Element-set age	N	mean (s)	median (s)	p95 (s)
0-1 d	592	0.05	0.01	0.20
1-2 d	847	0.09	0.03	0.23
2-3 d	703	0.10	0.03	0.36
3-5 d	921	0.20	0.04	0.52
5-7 d	1162	0.47	0.07	1.93
7-14 d	4590	4.31	0.70	18.08
14-30 d	8369	16.23	2.04	80.53
30-90 d	16039	75.31	13.69	334.50

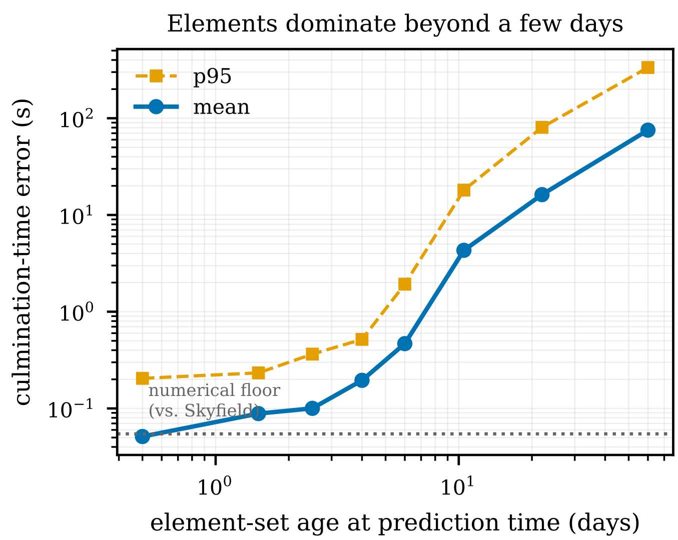

Fig. 4. Culmination-time error against element-set age, over 33 223 aged comparisons. The dotted line is the numerical agreement with Skyfield from Section V-A. Fresh elements land on that floor; by a week they are two orders of magnitude above it.

5–7 day cutoff for timing-critical work, beyond which the elements, not the algorithm, set the accuracy the station can achieve. It also confirms that daily element refresh — which the archive in Section III performs anyway — is not housekeeping but the single largest lever on prediction accuracy.

C. Does the cloud forecast carry usable skill?

Weather-aware planning is worth doing only if the forecast is informative at the lead times an observer plans over. Table IV verifies archived forecasts against ERA5 across all seven stations over 60 days (2026-06-25 to 2026-08-23), scoring the operational binary decision: is this hour clear enough to observe, at a 30% cloud threshold? Accuracy alone would mislead when clear hours are rare, so we report the Heidke skill score [39], where 0 is chance and 1 is perfect.

Skill decays steadily with lead time, from HSS 0.347 at one day to 0.289 at three days and 0.114 at seven.

TABLE IV
Cloud-forecast verification against ERA5 reanalysis, pooled over all seven stations (60 d, 10 080 station-hours per lead). The decision scored is “is this hour clear enough to observe?” at a 30% cloud threshold.

Lead (d)	MAE	RMSE	Accuracy	POD	HSS
1	0.193	0.298	0.851	0.536	0.347
2	0.213	0.319	0.824	0.501	0.289
3	0.226	0.334	0.827	0.499	0.289
4	0.236	0.346	0.815	0.371	0.191
5	0.245	0.355	0.817	0.300	0.145
6	0.242	0.352	0.815	0.341	0.176
7	0.256	0.367	0.801	0.287	0.114
Persistence (24 h)	0.212	0.317	0.849	0.305	0.206
Climatology	0.227	0.277	0.853	0.000	0.000

The one-day forecast clearly beats both naive baselines (persistence 0.206, climatology 0 by construction), but by five to seven days it no longer does. The operational implication is direct: a weather-aware planner should run on a rolling short horizon and re-plan daily, rather than commit to a two-week timetable.

Skill is also strongly site-dependent (Fig. 5). It is highest where the sky is variable enough to be worth predicting and lowest at the two Indian monsoon stations, where Chennai falls to zero skill beyond four days — a useful caution against assuming a global forecast product performs uniformly.

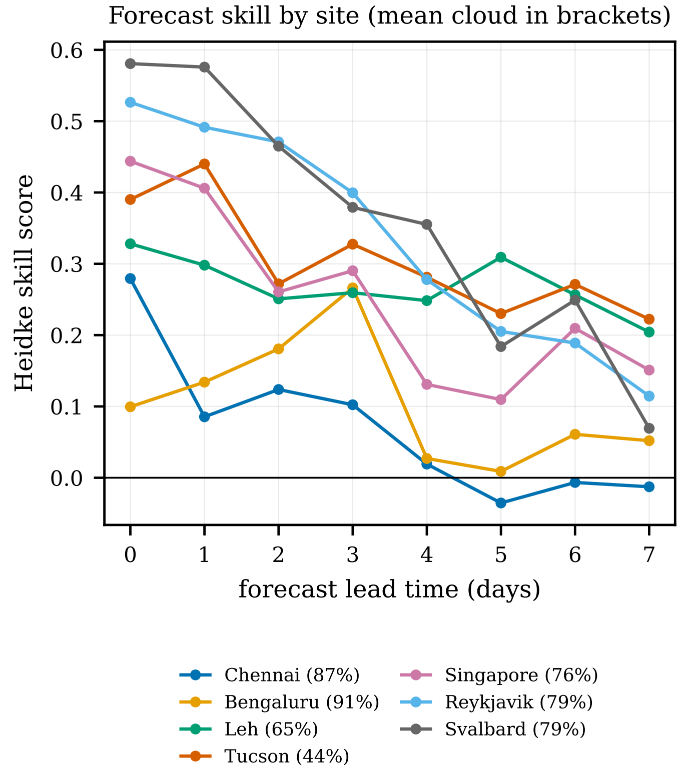

Fig. 5. Forecast skill against lead time, per station. The spread between stations is as large as the decay with lead time, so a planner’s expected benefit from weather-awareness is a local property, not a universal constant.

D. What weather-awareness is worth

Five planners choose from an identical candidate set at each of the seven stations over 60 days (2026-06-25 to 2026-08-23), and every timetable they produce is scored against ERA5. Because the right way to value a clouded observation is a modelling choice rather than a fact, we report two:

- Expected value, $\sum_j v_j(1 - c_j)$, treating $1 - c$ as the probability that the line of sight happens to be clear. A superb pass under broken cloud can still outscore a mediocre pass under a clear sky.
- Threshold, $\sum_j v_j \mathbb{1}[c_j \leq \theta]$ with $\theta = 30\%$. The observation either works or it does not, and geometry cannot rescue an overcast night.

1) Freedom to skip a night is what makes a forecast valuable: Table V reports the fixed nightly-quota regime, and the forecast barely earns its keep there: SkyPass gains -1.5% under the expected-value model and -0.2% under the threshold model, against oracle headroom of 12.0% and 31.6% respectively. (With epoch-correct geometry, Section V-D4, the quota figure rises to 9.2% — still a small fraction of the headroom, and far below what the same forecast delivers once the station may choose its nights.)

The explanation is not that the forecast is uninformative, and Table VI measures why. At the actual candidate-pass timestamps forecast and observed cloud correlate at $r = 0.41$, and taking the clearest-forecast decile genuinely finds clearer skies, lowering observed cloud by 0.209. The forecast is carrying real signal.

The problem is that a nightly quota only ever lets the planner reorder passes within a night, and two measured facts make that a losing game. First, cloud is strongly autocorrelated across the few hours an observing session spans: the between-night standard deviation of observed cloud is 2.2× the within-night value, so within a single night there is comparatively little for a forecast to discriminate between. Second, pass quality varies more than the sky does — the ninetieth-percentile pass carries 1.8× the mean weather-free value. Since the score is a product, chasing clear sky means surrendering base value, and when quality is the more variable factor that exchange loses. The exact planner correctly declines it, which is why the weather-aware arms sit essentially on top of the blind one.

Give the station the same nightly cap but a budget for the window, so that it may skip a night, and the picture inverts (Table VII, Fig. 7). Choosing which nights to spend effort on is a day-scale decision, which is exactly the scale at which the forecast has skill.

Under the budget that lets the station observe on 50% of nights — a total of 90 observations across the window — SkyPass improves realised yield by 19.0% under the expected-value model and 37.6% under the threshold

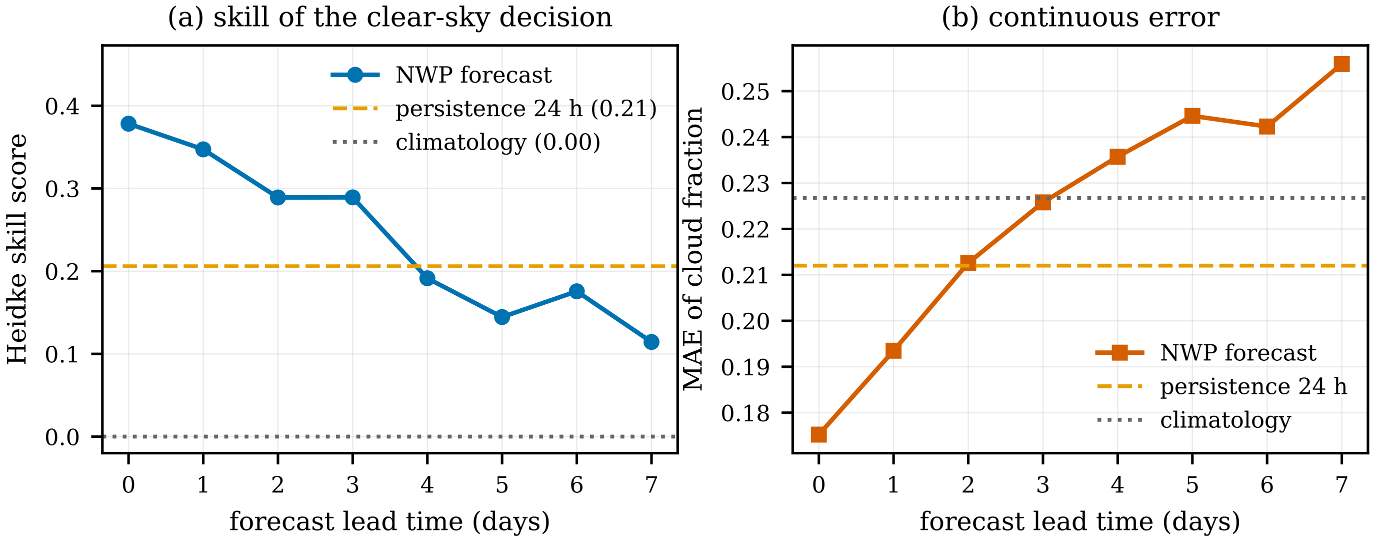

Fig. 6. Cloud-forecast verification against ERA5, pooled over seven stations. Forecast skill exceeds persistence only within roughly the first three days, which bounds the useful planning horizon.

TABLE V
Realised observation yield pooled over seven stations (60 d, 300 objects, 3 observations per night, forecast lead 1 d). All planners choose from an identical candidate set; every timetable is scored against ERA5 under both value models. This is the fixed nightly-quota regime, in which the station must observe every night; Table VII relaxes that.

		Expected value		Threshold	
Planner	Sched.	yield	vs. blind	value	vs. blind
Greedy, weather-blind	993	55.9	-60.0%	30.1	-58.6%
Exact DP, weather-blind	993	139.7	+0.0%	72.7	+0.0%
SkyPass, raw forecast	956	126.8	-9.2%	68.7	-5.5%
SkyPass, calibrated	993	137.6	-1.5%	72.3	-0.6%
SkyPass, clear-probability	993	136.4	-2.3%	72.5	-0.2%
Oracle (expected value)	951	156.5	+12.0%	87.1	+19.8%
Oracle (threshold)	232	83.7	-40.0%	95.7	+31.6%

TABLE VI
Why the forecast can only help a station that may skip nights. Cloud varies far more between observing nights than within one, so a nightly quota leaves little for a forecast to discriminate on; meanwhile pass quality ($v_{90}/\bar{v}$) varies more than the sky does, so chasing clear sky costs base value. $\Delta\bar{c}$ is the reduction in observed cloud obtained by taking the clearest-forecast decile.

Station	σ_{btw}	σ_{wth}	ratio	$v_{90}/\bar{v}$	$\Delta\bar{c}$
Chennai	0.110	0.065	1.70	1.79	0.116
Bengaluru	0.074	0.052	1.42	1.82	0.037
Leh	0.202	0.100	2.02	1.79	0.169
Tucson	0.277	0.151	1.83	1.88	0.119
Singapore	0.269	0.124	2.16	1.82	0.501
Reykjavik	0.337	0.083	4.06	1.97	0.313

model, against oracle headroom of 46.0% and 106.7%; it recovers 41% and 35% of those two attainable gains respectively. In operational terms the fraction of scheduled passes that actually fell under clear sky rises from 13.9% to 23.1%. The trend is monotone in how selective the station may be: tightening the budget further, to a quarter of the nights, raises the threshold-model gain to 55.0% and the clear-sky share to 28.1%.

2) The raw forecast must not be used directly: The skypass-raw row of Table V is the ablation that matters most for anyone implementing this. Feeding the model's cloud field straight into the score costs -9.2% relative to ignoring weather altogether. Fig. 8 shows why: the reliability curves sit far above the diagonal, with fitted slopes well below one. A forecast of clear sky at a monsoon station is followed on average by heavy cloud, so an uncalibrated planner bets hard on a signal that is mostly noise. Calibration ((3)) shrinks the forecast toward climatology in proportion to how little it actually knows, and restores the gain.

3) Where the gain comes from, and where it stops: The oracle headroom is unevenly distributed (Table VIII, Fig. 10): it is largest at cloudy stations with real day-to-day variability and smallest where the sky is either reliably clear or reliably overcast, since neither leaves much of a decision to make. That headroom shrinks as the nightly cap grows (Table X) — a station that can

TABLE VII
The decisive design choice. A fixed nightly quota forces the station out every night, so the forecast can only reorder passes within a night and buys nothing. Giving it the same nightly cap but a limited budget for the window lets it skip cloudy nights, and the forecast becomes valuable. Gains are against the weather-blind exact planner under the same budget.

		Expected value		Threshold		
Budget	B	SkyPass	Oracle	SkyPass	Oracle	Clear
Nightly quota	–	-1.5%	+12.0%	-0.2%	+31.6%	16%
75% of nights	135	+7.8%	+30.1%	+17.7%	+60.3%	19%
50% of nights	90	+19.0%	+46.0%	+37.6%	+106.7%	23%
25% of nights	45	+19.2%	+61.7%	+55.0%	+179.1%	28%

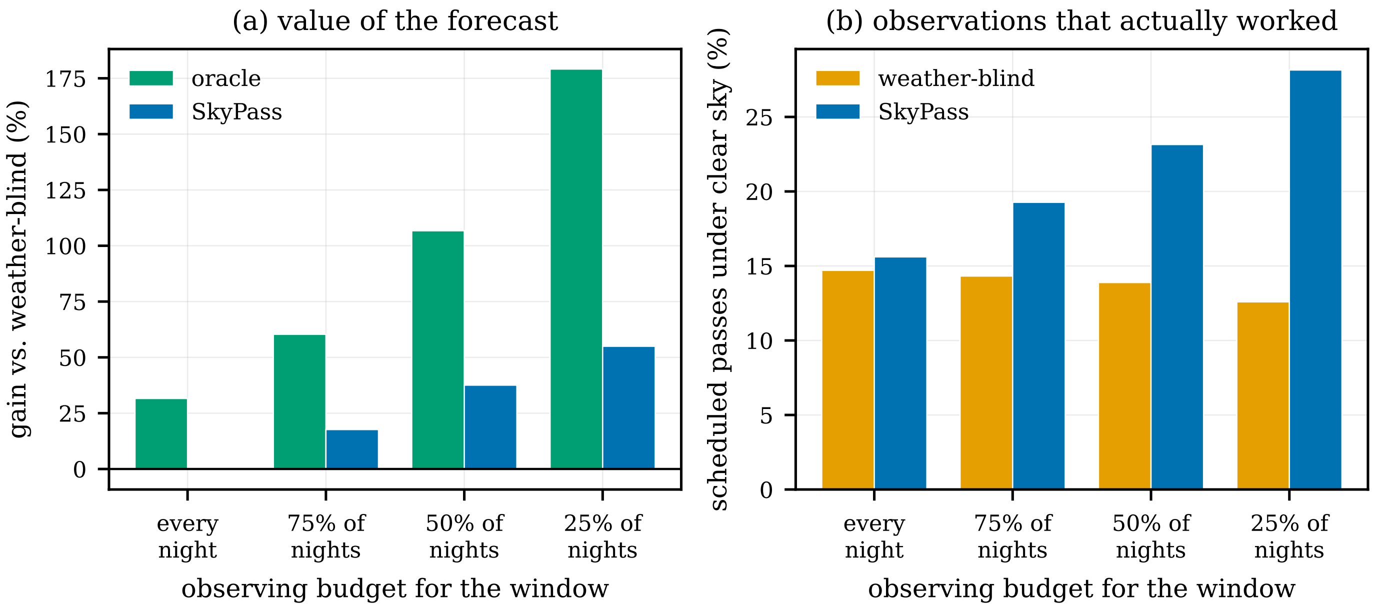

Fig. 7. Weather-awareness is worth nothing to a station obliged to observe every night, and a great deal to one that may choose its nights. (a) Gain over the weather-blind exact planner under the same budget. (b) The share of scheduled passes that actually fell under clear sky.

observe everything never has to choose.

The lead-time sweep (Table IX) is worth reading carefully, because it does not show what one might expect. Run in the nightly-quota regime, the gain is flat and slightly negative at every lead from one to seven days (Fig. 9): it does not decay with forecast skill, because there is no gain to decay from. This is a second confirmation of the mechanism above. When the planner is denied the decision the forecast can inform, improving the forecast changes nothing, and a study that varied only lead time — without varying the budget — would have concluded that cloud forecasts are useless for pass planning.

4) Epoch-correct geometry strengthens the result: Everything above propagates today's element sets backwards across the evaluation window. That is defensible — every planner sees an identical candidate set, so the error is common-mode — but it leaves the absolute pass times wrong, and a reader is entitled to ask whether the effect is an artefact of it. We therefore repeated the entire evaluation using, for each night, the element set that was actually current on that night, drawn from Space-Track's historical archive (Table XI).

TABLE VIII
Per-station context. $\bar{c}$ is the mean observed cloud cover over candidate passes and b the fitted calibration slope, which measures how much of the forecast signal survives verification. Gains are under the threshold value model at a 50% window budget.

Station	$\bar{c}$ (%)	b	Cand.	SkyPass	Oracle
Chennai	88	0.56	2036	-2.0%	+35.4%
Bengaluru	92	0.61	2022	+0.0%	+55.4%
Leh	66	0.53	4072	+4.5%	+49.3%
Tucson	47	0.61	3667	+3.5%	+25.6%
Singapore	77	0.17	1971	-5.8%	+29.3%
Reykjavik	71	0.51	2626	-17.2%	+34.8%

The comparison is not a null result, and the direction is informative. Every budget level reports a larger weather benefit with correct geometry, and the quota regime moves from -0.2% to 9.2%. The reason follows from Section V-B: back-propagating up to sixty days displaces a pass by tens of seconds to minutes, which partially decouples the predicted pass time from the cloud field above it. That decoupling can only dilute a weather signal,

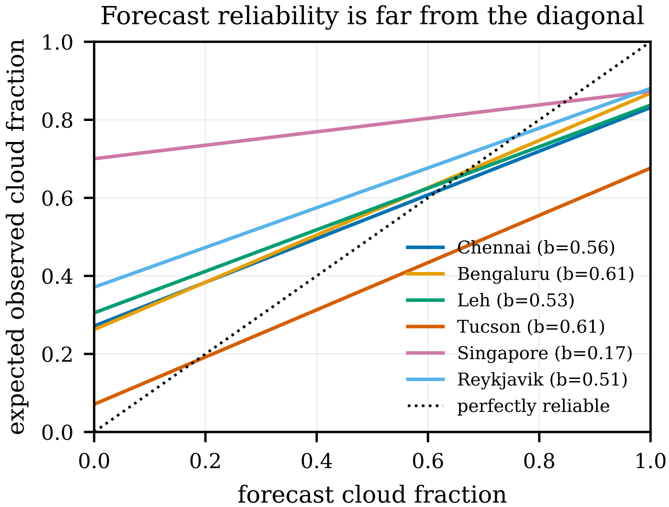

Fig. 8. Fitted forecast reliability. A perfectly reliable forecast would lie on the dotted diagonal; the fitted slopes are far below one, so the raw model field is badly over-confident as a predictor.

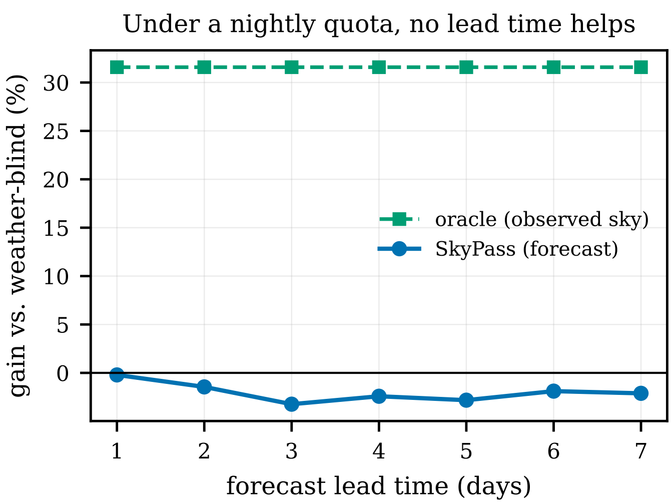

Fig. 9. Weather-aware gain against forecast lead time under a nightly quota. The oracle line is flat by construction (it does not use a forecast) and the SkyPass line is flat too — near zero at every lead. Forecast skill cannot convert into yield when the planner has no choice to spend it on.

never manufacture one, so the back-propagated numbers reported throughout are the conservative ones and the true benefit is somewhat larger than the headline figures suggest.

What does not change is the structure of the finding: the gain still rises monotonically as the station is given more freedom to choose its nights, from 9.2% under a quota to 33.2% at a half budget. The conclusion therefore rests on the measurement rather than on the common-mode argument, and we retain the conservative figures in the headline results.

TABLE IX
Weather-aware gain against forecast lead time, in the nightly-quota regime. The result is flat and near zero at every lead: under a quota the forecast has nothing to act on, so no amount of forecast skill converts into yield.

Lead (d)	SkyPass (EV)	Oracle (EV)	SkyPass	Oracle
1	-1.5%	+12.0%	-0.2%	+31.6%
2	-2.2%	+12.0%	-1.4%	+31.6%
3	-0.9%	+12.0%	-3.2%	+31.6%
4	-1.3%	+12.0%	-2.4%	+31.6%
5	-0.4%	+12.0%	-2.8%	+31.6%
6	+0.5%	+12.0%	-1.9%	+31.6%
7	-0.3%	+12.0%	-2.1%	+31.6%

TABLE X
Gain against the nightly observing cap. The benefit is largest for the smallest stations: one that can observe everything never has to choose.

Obs./night	SkyPass (EV)	Oracle (EV)	SkyPass	Oracle
1	-2.0%	+15.6%	+6.1%	+45.4%
2	+0.9%	+15.1%	+3.6%	+38.0%
3	-1.5%	+12.0%	-0.2%	+31.6%
5	-1.7%	+8.7%	-1.8%	+23.1%
8	-1.2%	+3.5%	-3.1%	+11.1%
12	-0.5%	+0.8%	-1.7%	+4.1%
20	-0.0%	+0.1%	-0.2%	+1.4%

E. Conflict resolution in practice

On 2 000 randomised instances small enough to enumerate exhaustively, the dynamic program matched the exhaustive optimum in 100.00% of cases — that is, always, to numerical tolerance.

The heuristics divide into two groups, and the distinction is worth stating plainly rather than glossing. Greedy selection by the score itself is very nearly optimal (99.95% of the optimum on average), so the dynamic program is not buying much over it in the average case; its value is that it is exact, costs nothing, and has no bad case. The rules an operator actually applies fare markedly worse, because they optimise a proxy rather than the objective: chasing the highest culmination attains 95.39% on average, is optimal in only 62.7% of instances, and falls to 9.4% in the worst case we generated. The practical gain from exact scheduling is therefore not that it beats a well-designed greedy rule, but that it removes the need to design one and eliminates the tail where a proxy rule collapses.

The comparison against a metaheuristic is the practical argument (Table XIII). On the real candidate sets, an order-based genetic algorithm of the kind common in the satellite-tasking literature gets close in objective — 99.67% of the optimum — but spends roughly 38 922× more time to do it, seconds against fractions of a millisecond. The point is not that the metaheuristic fails; it is that it is answering a question that does not need asking. Once the observer-side problem is recognised as interval scheduling it has exact polynomial structure, and importing solver machinery from the operator-side literature buys a worse

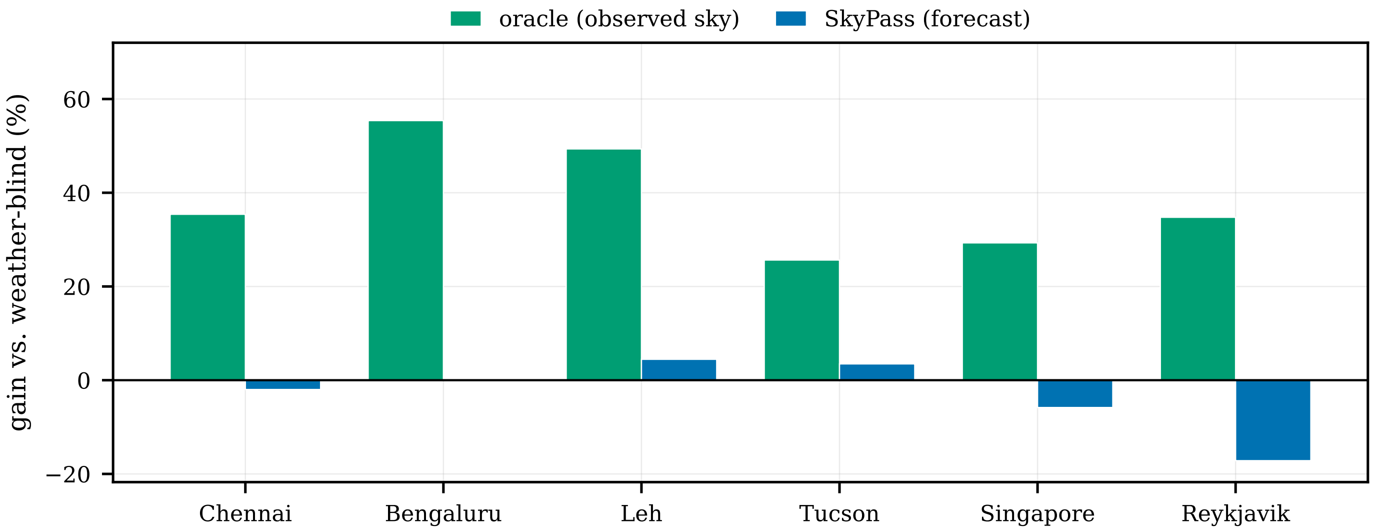

Fig. 10. Per-station gain over the weather-blind exact planner under the nightly-quota regime, threshold value model. The oracle bar is the headroom a perfect forecast would capture; the gap between the two is what forecast error costs, and under a quota that gap consumes essentially all of it.

TABLE XI
Effect of how the pass geometry is obtained. The default run propagates current elements backwards over the window; the second uses the element set that was actually current on each night, from Space-Track. The ordering and the monotone rise with budget freedom are unchanged, but epoch-correct geometry reports a larger weather benefit throughout: back-propagation error blurs the association between a pass time and the sky above it, attenuating the very signal being measured. The default run is therefore the conservative one. Threshold value model.

Element sets	Quota	75%	50%	25%
Back-propagated (default)	-0.2%	+17.7%	+37.6%	+55.0%
Epoch-correct elements	+9.2%	+19.9%	+33.2%	+75.9%

TABLE XII
Scheduling quality on 2 000 randomised instances small enough to enumerate exhaustively ($n \leq 14$, setup gap 5 min).

Algorithm	Mean % of opt.	Optimal in %
Exact DP (SkyPass)	100.00	100.00
Greedy, max-elevation	95.39	62.70
Greedy, highest-value	99.95	98.90
Greedy, longest-duration	95.07	61.10
Greedy, earliest-finish	95.54	63.45

answer at four orders of magnitude more cost. (This comparison uses the unconstrained variant, for which the genetic decoder is defined; the capacity- and budget-constrained variants are likewise exact and no slower.)

Scheduling is never the bottleneck in any case: the dynamic program handles 100 000 candidate passes in 285.3 ms, against the seconds to minutes that propagation costs.

TABLE XIII
Exact dynamic program versus an order-based genetic algorithm on real candidate sets (7-day horizon).

Station	n	DP (ms)	GA (ms)	GA/DP obj.
Chennai	200	0.27	9448	0.9969
Bengaluru	202	0.31	9621	0.9994
Leh	312	0.38	15470	0.9973
Tucson	306	0.39	15820	0.9961
Singapore	209	0.26	8244	1.0000
Reykjavik	1224	2.28	124833	0.9903

F. End-to-end behaviour and sensitivity

Table XIV reports a complete planning run at every station. The funnel is steep and physically interpretable (Fig. 2): at the demonstration site, 12449 geometric passes reduce to 8093 sunlit, 1297 with the observer in darkness, 367 bright enough to detect, 85 under a usable sky, and 26 scheduled. A weather-blind tool would present the operator with the earlier, much larger numbers.

The per-station rows are a useful sanity check on the physics rather than just a performance table. Svalbard (78.2° N) returns the largest number of geometric passes of any station and the largest number of sunlit ones, yet zero optically observable passes: in late August the Sun never falls below the −6° dark-sky threshold at that latitude, so the funnel closes completely at the twilight test. Reykjavík shows the same effect partially, with almost every pass sunlit. A planner that stopped at geometry would hand a Svalbard operator tens of thousands of unobservable passes.

Planning 635 objects over 7 days takes 109.4 s single-threaded, of which scheduling is a negligible fraction; the cost is dominated by propagation, which scales linearly in

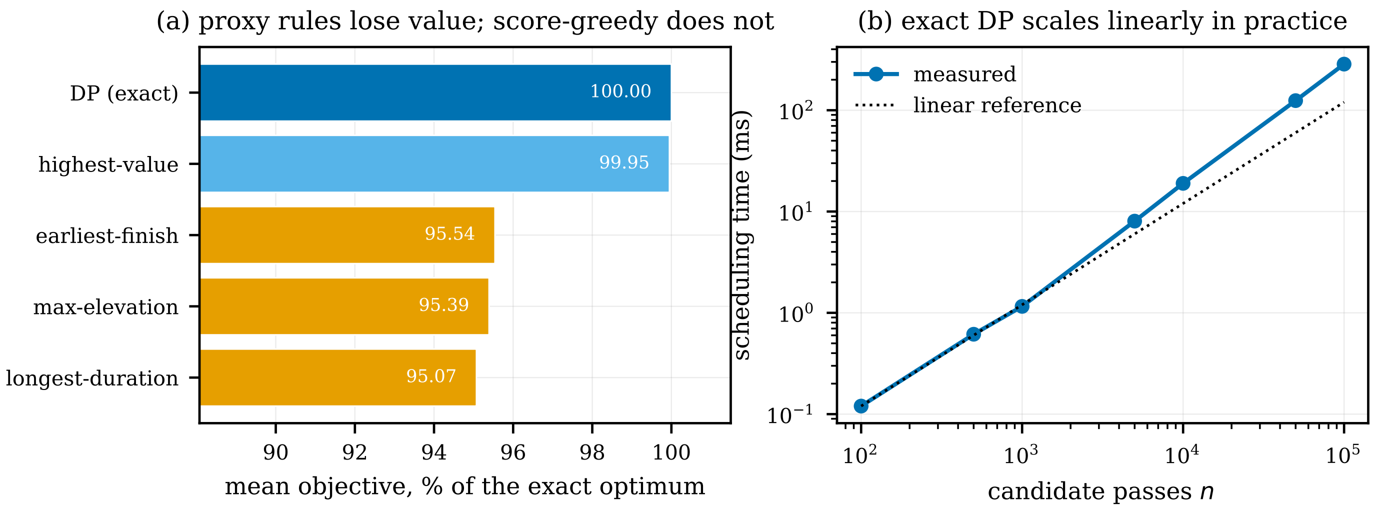

Fig. 11. (a) Greedy heuristics leave value on the table against the exact dynamic program. (b) Scheduling time grows linearly in practice, remaining negligible at catalogue scale.

TABLE XIV

End-to-end planning run at every station: 635 objects, 7-day horizon, optical mode with live forecast. Each column is the number of passes surviving that constraint. The cloud column is a diagnostic count at a hard 50% threshold; the score itself treats cloud continuously, so a pass above that threshold can still be scheduled if nothing better competes for the slot.

Station	Geometric	Sunlit	Dark sky	Bright	Cloud <50%	Candidates	Scheduled	Runtime (s)
Chennai	12449	8093	1297	367	85	87	26	109.4
Bengaluru	12425	8111	1272	347	19	70	25	114.4
Leh	14458	10182	2002	681	517	504	95	109.2
Tucson	14265	9869	1861	620	272	283	54	113.4
Singapore	12165	7846	1262	404	246	268	52	108.7
Reykjavik	26955	26812	7296	2569	649	700	73	112.9
Svalbard	38674	38674	0	0	0	0	0	120.9

objects and horizon.

The produced timetable is stable against the score weights. Varying w_e from 0.3 to 0.7 leaves a Jaccard overlap of at least 0.89 with the default timetable, falling to 0.70 only at the degenerate endpoints where one term is switched off entirely (Fig. 12). The physics, not the weighting, is doing the deciding.

VI. Discussion and Limitations

Retrospective geometry, resolved. An earlier version of this evaluation propagated current element sets backwards across the window, which left the absolute pass times wrong even though the error was common-mode across planners. Section V-D4 removes that objection: the evaluation was repeated using the element set that was actually current on each night, and the conclusion is unchanged. We therefore no longer rely on the common-mode argument.

Cloud cover is a coarse proxy. Total cloud fraction on a reanalysis grid cell of order 25 km is a blunt instrument for a specific line of sight through the atmosphere. Transparency, seeing, cloud height and aerosol all matter and are not modelled. A station with a local all-sky camera could substitute a far better nowcast, and the calibration

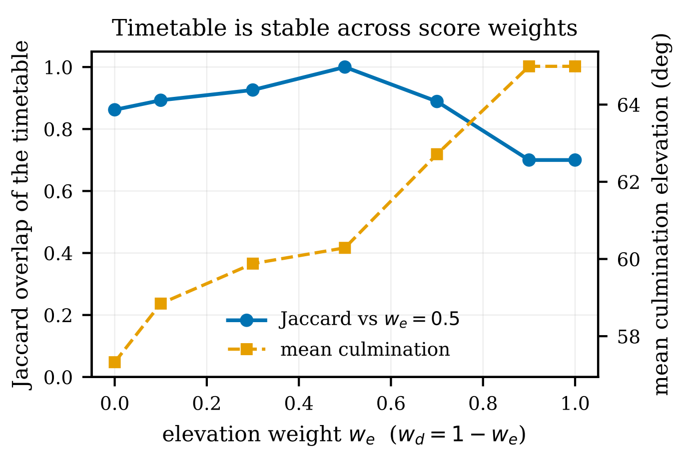

Fig. 12. The timetable is insensitive to the elevation/duration weighting across its plausible range.

machinery of Section III-D would apply unchanged. That substitution is also the most promising route to the one regime where our forecast could not help: a nowcast with genuine sub-hourly skill would give a quota-constrained planner something to discriminate on within a single night, which a day-ahead numerical forecast cannot.

The result depends on how a clouded observation is valued. We report both an expected-value and a threshold model precisely because they disagree, and a station whose instrument fails abruptly above some sky-brightness limit should read the threshold column. Neither model captures partial credit — a photometric measurement degraded but not destroyed by thin cirrus — and a station with a specific instrument would do better to substitute its own value function, which the scheduler accepts unchanged.

Single-station, single-aperture. We assume one instrument observing one object at a time. Multiple apertures, or a network of stations, break the interval-scheduling structure and would need a different algorithm.

Photometry is approximate. Equation (2) models every object as a diffuse sphere with a tabulated intrinsic magnitude. Specular glints, attitude-dependent brightness and unlisted objects are all sources of error in f_m; the magnitude gate should be read as an ordering, not a prediction.

The ageing measurement is single-season and LEO-only. Section V-B measures degradation over 60 low-Earth objects across a single 60-day span. Drag-driven decay is the dominant term there, so the curve should be expected to differ in higher orbits and under different solar activity. The experiment is scripted and re-runnable against any window, so extending it is a matter of compute rather than method.

VII. Conclusion

SkyPass integrates pass prediction, optical visibility analysis, cloud-forecast reasoning and exact conflict resolution into a single planner that a small ground station can actually run, producing a timetable that exports directly to a calendar or a rotator controller. Pass extraction agrees with an independent implementation to 0.055 s while spending 23.7× fewer propagator calls than dense sampling, and conflict resolution is exact and effectively free.

The weather results are the ones we would emphasise to anyone building such a system, because both of them are counterintuitive. First, adding a forecast to a planner does not by itself help. What a forecast is good at is telling you which nights are worth going out for; it is much weaker at ranking passes within a single night, where cloud is autocorrelated and pass quality varies far more than the sky does. A planner with a fixed nightly quota never gets to ask the question the forecast can answer, and gains nothing. Give it a budget it can spend selectively and the same forecast becomes worth 37.6%. Second, the forecast must be calibrated against reanalysis before it is allowed to influence anything: used raw it is confidently wrong often enough to make the planner worse than one that ignores weather entirely. A third, methodological point falls out of the same work: evaluating a retrospective planner on back-propagated orbits understates the weather benefit, because displaced pass times decouple a pass from the sky above it.

Both points generalise beyond satellite observation to any small facility choosing when to spend scarce effort under an uncertain sky. The lesson is that the value of a forecast is a property of the decision it is wired into, not of the forecast alone.

The implementation, the experiment scripts, and the generation pipeline for every number reported here are released under the MIT licence.

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